

PERFORMANCE TESTING (CONTINUATION)

OptiCrop: Smart Agricultural Production Optimization Engine

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Team ID	SWTID-2026-7253
Team Size	5
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Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value
1	Response Time (Avg)	less than 2 seconds[cite: 24]	1.18 seconds[cite: 24]
2	Response Time (Max)	5 seconds[cite: 24]	2.86 seconds[cite: 24]
3	Throughput (Req/sec)	-	1487 req/sec[cite: 24]
4	Error Rate	-	0.32%[cite: 24]
5	CPU Utilization	-	63%[cite: 24]
6	Memory Utilization	-	56%[cite: 24]

Step 4: Observations & Analysis

Key Findings:

- The system responded within the acceptable time range for both average and maximum response time[cite: 24].
- Throughput was stable and the system handled concurrent user load efficiently[cite: 24].
- Error rate remained well below 1%[cite: 24].

Bottlenecks Identified:

- High database query time during peak concurrent requests from regional farming sectors[cite: 24].
- Some delay observed in loading dashboard analytics due to heavy multi-variable agricultural data processing[cite: 24].

Optimization Steps Taken:

- Optimized SQL queries and added indexing on frequently accessed soil-climate parameter tables[cite: 24].
- Implemented caching for dashboard data to reduce response time for crop profiles[cite: 24].
- Enabled connection pooling to improve database transaction performance under load[cite: 24].

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):[cite: 24]

[Screenshot 1 Placeholder: JMeter Summary Report for Crop Prediction Endpoint][cite: 24]

[Screenshot 2 Placeholder: Response Time Graph Under Concurrent Soil Parameter Ingestion][cite: 24]